

Ziad Atef

Data Science & Machine Learning Engineer

ziad.atef.yehia@gmail.com

+20 10 68131634

Egypt

linkedin.com/in/ziad-atef-747a54351

PROFESSIONAL SUMMARY

Computer Science & AI Engineer focused on Data Science and Machine Learning, with hands-on freelance experience delivering customer retention models and automated sales analytics for real clients. Experienced in Python, SQL, data analysis, data visualization, machine learning, and computer vision.

Built a customer retention classification solution that improved prediction accuracy from a **78% baseline to 90%** with XGBoost, achieving a **0.94 F1-score**. Delivered automated Arabic sales reporting solutions for **2 e-commerce clients**, transforming raw business data into actionable insights and client-ready PDF reports.

Currently completing DEPI's 6-month Data Science track and developing a YOLOv8-based graduation project for real-time person detection and occupancy estimation.

WORK HISTORY

Freelance Data Science & Machine Learning Engineer

2025 – 2026

Independent Freelance Client Projects

Egypt

- **Built and delivered** a customer retention classification pipeline for a client developing a digital ordering platform for a restaurant brand, transforming **3,000 raw order-level records into 786 unique customer profiles**.
- **Engineered the retention target** from historical customer order behavior when a direct retention label was unavailable, applying data preprocessing and preventing target leakage during dataset preparation.
- **Addressed severe class imbalance** using SMOTE and compared Logistic Regression, Decision Tree, Random Forest, and XGBoost using Stratified K-Fold cross-validation and GridSearchCV.
- **Achieved 90% prediction accuracy with XGBoost**, improving the **78% Logistic Regression baseline by 12 percentage points**, with a final **0.94 F1-score**, and delivered feature-importance visualizations and a final client report.
- **Delivered automated data analysis and visualization solutions for 2 e-commerce clients**, transforming raw sales data into structured Arabic PDF reports using Python, Pandas, Seaborn, and Plotly.
- **Identified a high-performing city that drove 45.7% of total revenue** for one client, directing marketing budget toward the strongest-performing region.
- **Found a product category that drove 61.3% of total sales** and measured a **0.80 correlation** between customer rating and sales value, supporting marketing and product prioritization.
- **Flagged an 85% revenue decline month-over-month** (from a January peak of 691,300 EGP to an April low of 106,200 EGP) for a second client, enabling early business intervention.
- **Analyzed 3.38M EGP in revenue across 100 orders** and found a product category that drove **79% of total revenue**, translating findings into marketing and inventory recommendations.
- **Automated PDF report generation** using Python, WeasyPrint, and ReportLab, producing structured, client-ready Arabic business reports.

EDUCATION

Bachelor's Degree in Computer Science and Artificial Intelligence

2023 – 2027

Assiut National University – Faculty of Computers & Artificial Intelligence

Assiut, Egypt

Expected Graduation: 2027

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, Pandas, NumPy

Machine Learning: Scikit-learn, XGBoost, Keras, Machine Learning, Feature Engineering, Model Evaluation, Cross-Validation, SMOTE, GridSearchCV

Computer Vision: YOLOv8, Computer Vision, Object Detection

Data Visualization & Reporting: Matplotlib, Seaborn, Plotly, Excel, Data Visualization, Data Engineering, Arabic Reporting, WeasyPrint, ReportLab

Tools: Flask, MLflow, Hugging Face

SOFT SKILLS

Analytical, Problem-Solving, Communication, Collaboration

PROJECTS

Smart Campus Space Management System — Graduation Project

- **Developing** a YOLOv8-based Computer Vision system for real-time person detection and lecture hall occupancy estimation to support smart campus space management.